

HiGrEC: A Controlled Study of Structural Inductive Bias for Response Clarity Classification in Political Interviews

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Abstract

SemEval-2026 Task 6 (CLARITY) asks how a politician’s answer responds to an interview question: one of nine evasion types (Subtask 2), which fixes one of three clarity levels (Subtask 1). We report a controlled, pre-registered study of what limits single-pass classifiers on this task. A DeBERTa-v3-large classifier, improved one measured change at a time, reaches 0.384 multi-reference macro-F1 on Subtask 2 per model and 0.405 as a ten-seed system with a one-parameter logit adjustment. Every attempt to improve the *decision* rather than the model (re-ranking, three hierarchies, rebalancing losses, model soups) failed, and error analysis points to the model, not label noise, as the bottleneck. Replacing only the backbone with Qwen3-8B fine-tuned with LoRA, with data, input, loss and selection unchanged, improves Subtask 2 by 0.092 and Subtask 1 by 0.095 on all ten paired seeds and reaches 0.543 / 0.746 as a system (+0.136, 95% CI [0.054, 0.224]). The gain is spread across classes rather than concentrated on the label boundaries we predicted. A longer schedule is adopted on a held-out slice of the training data; training on all data adds no distinguishable gain. All scores are on the development set, since test labels were not released.

1 Introduction

Politicians often answer the question they would have liked to be asked. Whether an answer gives the requested information, gives it only implicitly, or avoids it by dodging, deflecting or declining is a long-standing object of study in political communication (Bull, 1994), and Thomas et al. (2024) turned it into a classification task with a two-level taxonomy of response clarity and evasion. SemEval-2026 Task 6 (Thomas et al., 2026) made this a shared task: Subtask 2 asks for one of nine evasion types, and Subtask 1 for one of three clarity levels that the nine types determine.

The competition was won by multi-call large language model (LLM) pipelines (Shi et al., 2026; Peerachaidecho and Rutherford, 2026), while the organisers report that fine-tuned encoders stopped at about 0.50 macro-F1 on Subtask 2 regardless of scale or ensembling (Thomas et al., 2026). A leaderboard does not say *why* encoders stop there. The candidate explanations make different predictions: the labels may be too noisy to learn (then annotators would disagree where models fail), the metric may reward something the model is not optimising (then a better decision rule would help), the model may lack the knowledge needed to read an answer against its question (then a larger pretrained model would help under the same protocol), or the label meanings may not be learnable from 3,448 examples (then supplying definitions would help).

We study these explanations with a deliberately controlled single-pass classifier. Every change is made one at a time against a named control, evaluated on ten paired seeds, with hypotheses and predictions written down before each run and no choice made on the development set (§2). Our contributions are:

- A carefully built DeBERTa-v3-large system (§4) whose two real improvements (training length and the full journalist question) are separated from a series of documented negative results on the decision layer: re-ranking, hierarchical models, rebalancing losses, per-class thresholds and model soups.
- Evidence that the remaining gap lies in the model rather than in label noise or the decision rule (§6).
- A one-change test of the knowledge explanation (§5): replacing the backbone with Qwen3-8B-Base fine-tuned with LoRA (Qwen Team, 2025; Hu et al., 2022) improves both subtasks on all ten seeds, and as a ten-seed system

raises Subtask 2 from 0.405 to 0.543 and Subtask 1 from 0.648 to 0.746 on the development set.

- An analysis of where that gain falls, which refutes our own pre-registered prediction that it would concentrate on the boundaries between commitment levels (§6).

The test labels were never released and the evaluation server has closed, so all comparisons are on the 308-item development set; we report seed spreads, paired differences and bootstrap intervals throughout to state each claim at the resolution this set supports.

2 Task, Data and Evaluation

Task. SemEval-2026 Task 6 (Thomas et al., 2026) uses the question–answer pairs of Thomas et al. (2024), taken from U.S. presidential interviews. Each item is a triple: one sub-question (median 15 tokens), split out of the journalist’s full turn (median 62 tokens), and the president’s entire answer (median 266 tokens). The label states how *that sub-question* was answered. Subtask 2 (S2) has nine evasion types; Subtask 1 (S1) has three clarity levels that are a fixed function of them: Clear Reply (Explicit; 30.5% of train), Ambivalent (Implicit, Dodging, General, Deflection, Partial/half-answer; 59.2%) and Clear Non-Reply (Declining to answer, Claims ignorance, Clarification; 10.3%). Like the dataset paper and the two top-ranked systems (Shi et al., 2026; Peerachaidecho and Rutherford, 2026), we predict the 9-way label and derive the 3-way label from the taxonomy.

Data. Train has 3,448 items with one label each and an annotator identifier (three annotators, one per item). Dev has 308 items labelled by three annotators: 125 are unanimous, 150 split 2–1 and 33 three ways, and 59.4% carry more than one distinct label. Test has 237 items with two annotators each; its labels were never released and the evaluation server has closed, so every comparison in this paper is on dev. Two properties shape the modelling. First, 69% of training rows share their answer with another sub-question, and 71.7% of those shared answers carry different labels, so the model must attribute parts of a long answer to the right sub-question. Second, answers are long: for 101 of the 308 dev items the sub-question and answer exceed 512 DeBERTa tokens, and for 20 the

Floor (dev)	S2		S1
	3 ann.	2 ann.	
Always majority (Explicit)	0.060	0.055	0.136
Uniform random (20 draws)	0.129	0.116	0.299
TF-IDF + logistic regression	0.257	0.243	0.445

Table 1: Trivial floors on dev. “2 ann.”: mean over the three two-annotator versions of the dev reference sets, the test regime. TF-IDF uses word 1–2-grams; its settings were chosen by 5-fold cross-validation on train.

full input exceeds 1,024. Test answers are much shorter (median 71 tokens against 266 on train and 362 on dev).

Metric. Both subtasks are scored by macro-F1 over all declared classes; a class that is never predicted contributes $F_1 = 0$. S1 uses plain single-label macro-F1. S2 uses *multi-reference macro-F1*: with G_i the set of annotator labels of item i , a prediction $\hat{y}_i$ is a true positive for class $\hat{y}_i$ if $\hat{y}_i \in G_i$, a false positive if not, and a miss charges one false negative to every member of G_i . If every prediction is acceptable ($\hat{y}_i \in G_i$ for all N items), there are no false positives or false negatives, every class predicted at least once has $F_1 = 1$, and

$$\text{macro-F1} = \frac{|\{\hat{y}_1, \dots, \hat{y}_N\}|}{9}. \quad (1)$$

Which acceptable label is named contributes nothing: a uniformly random member of each dev reference set scores 1.000 on five of five draws. The score therefore rewards two things, landing in the reference set (the *in-set rate*) and *coverage*, naming each class at least once. Since every class is worth one ninth of the score, a rare class such as Clarification, present in only 4 dev reference sets, counts as much as Explicit (115). The floors in Table 1 show the trade-off: always predicting Explicit lands in-set for 37.3% of items but scores 0.060, while uniform guessing lands in-set less often yet scores 0.129.

Evaluation protocol. The following protocol holds throughout the paper. *Epoch selection* uses the *train slice*, a fixed stratified 10% of train (345 rows, seed 12345) held out from training; dev is never used to choose anything. Every system claim and every final comparison rests on *10 paired seeds*, the same seed numbers for both arms, reported as mean $\pm$ standard deviation, with the per-seed difference, its t statistic and the number of seeds that improve. Variants are first screened on three seeds

against a bar fixed in advance (+0.015 with at least two of three seeds up). A *single model* is one seed with argmax and no tuning on dev. A *system* averages the seeds’ probabilities and applies logit adjustment (Menon et al., 2021):

$$\hat{y}(x) = \arg \max_c \frac{\bar{p}(c | x)}{\pi_c^\tau}, \quad (2)$$

where $\bar{p}$ is the mean of the seeds’ probabilities, π_c the frequency of class c in train, and τ the only fitted parameter. Because τ is fitted on dev, it is chosen by 5-fold nested cross-validation (grid -1 to 2 in steps of 0.05): each fold’s τ maximises S2 macro-F1 on the other four, and the pooled held-out predictions are scored. We report the first cross-validation split and, where given, the mean over ten splits; S1 is derived from the system’s final 9-way prediction. Two systems are compared by a bootstrap over dev items with the seeds fixed (2,000 resamples; Efron and Tibshirani, 1993), which understates training-seed noise. Because test has two annotators, we also rescore dev against each of its three two-annotator reference sets and report the mean. Finally, the hypotheses and predicted outcomes of every experiment were recorded in the experiment log before it ran, and results are reported against them.

3 Related Work

Response clarity and evasion. Political science has long described how politicians reply, partly reply or do not reply to interview questions (Bull, 1994). Thomas et al. (2024) turned this into an NLP task. They collected 287 U.S. presidential interviews (2006–2023), used ChatGPT to split multi-part questions into single question–answer pairs, and labelled each pair with a two-level taxonomy: three clarity classes over nine evasion leaves. Agreement among their three annotators on the 317 items they all labelled was moderate (Fleiss’ κ 0.644 for clarity, 0.48 for evasion), and they released every annotation alongside the adjudicated label. For both prompting and tuning, predicting the nine-way evasion label and mapping it up the hierarchy gave better clarity scores than predicting clarity directly.

The SemEval-2026 shared task (Thomas et al., 2026) reused this data with a new 237-item test set. It drew 124 registered teams, 41 of which submitted to the leaderboard, with 946 valid S1 runs and 539 valid S2 runs. The organisers’ baseline, a

fine-tuned Llama-70B, scored 0.82 (S1) and 0.57 (S2) macro-F1 on test. The organisers concluded that LLM prompting and hierarchical use of the taxonomy were the most effective strategies, and that systems with any hierarchical decomposition beat direct nine-way inference. Fine-tuned encoders levelled off near 0.50 on test S2 regardless of scale, ensembling or architectural additions. The organisers also singled out chain-of-thought distillation into 4B–14B open-weight models as a promising direction. Test labels have not been released, so all our comparisons with these systems use their development-set scores.

Top systems. TeleAI (Shi et al., 2026) ranked first on both subtasks (test 0.89 S1, 0.68 S2). Their pipeline runs three confidence-routed chain-of-thought stages on DeepSeek-V3. A gate outputs confident non-replies directly, sends uncertain ones to a nine-way review, and passes the rest to a six-way stage whose prompts include boundary examples mined from the training-set confusion matrix. The pipeline averages 1.94 model calls per item and reaches 0.617 (S2) and 0.812 (S1) on dev. In their ablation, the boundary few-shot examples gave the largest S2 gain (+0.097). In the same paper, Qwen2.5-7B fine-tuned directly reached 0.495 S2 on dev, level with single-step chain-of-thought prompting of DeepSeek-V3 (0.490). ChulaNLP (Peerachaidecho and Rutherford, 2026), second on S2 (0.61 test), took the top-five labels from a fine-tuned DeBERTa-large and passed them to few-shot Kimi-K2 as its only allowed labels. On dev, this hybrid scored 0.52 S2, against 0.46 for the encoder alone.

Methods we build on. For class imbalance, logit adjustment (Menon et al., 2021) subtracts scaled log class priors from the logits after training. Balanced Softmax (Ren et al., 2020) applies a similar prior correction inside the training loss, and focal loss (Lin et al., 2017) down-weights well-classified examples; we tested these two together. Our multi-reference scoring and our analysis of contested items follow work that treats annotator disagreement as signal rather than noise (Plank et al., 2014; Uma et al., 2021). Our encoder is DeBERTa-v3-large (He et al., 2023). Our LLM classifier is Qwen3-8B-Base (Qwen Team, 2025), fine-tuned with LoRA (Hu et al., 2022) and used as a single-pass classifier, whereas the top pipelines rely on chain-of-thought prompting (Wei et al., 2022). Model soups (Wortsman et al., 2022), which

average the weights of fine-tuned models, were our attempt to fold a seed ensemble into one model.

Positioning. We do not propose a leaderboard system. Instead, we run a controlled, pre-registered study of single-pass classifiers that asks what limits a fine-tuned encoder on this task and how much of the gap to multi-call LLM pipelines a single fine-tuned LLM closes.

4 The Encoder Track

Our first system is a fine-tuned encoder. We built it one controlled change at a time against a frozen baseline, and we judged each change by comparisons paired by seed, under the protocol of §2.

Model and training. We fine-tuned all 435M parameters of DeBERTa-v3-large (He et al., 2023) as a 9-way classifier with Wolf et al. (2020). The [CLS] state passes through a dense tanh pooler and a new linear layer over the nine evasion labels, and Subtask 1 is read off the predicted label. The baseline input pairs the sub-question (segment A, at most 96 tokens) with the answer (segment B), cut from the right at 512 tokens. The final input puts “Sub-question: ... Full question: ...” in segment A, with up to 256 tokens for the journalist’s whole turn, and raises the context to 1,024 tokens. The sub-question comes first, so truncation never removes it. We added the full question because 69% of training rows share their answer with another sub-question, and a model that cannot see the other sub-questions cannot tell which part of the answer responds to which. Training used plain cross-entropy, AdamW (Loshchilov and Hutter, 2019) at a learning rate of 10^{-5} with layer-wise decay 0.95 (10^{-4} for the head), a cosine schedule with 10% warm-up, batch size 16, and bf16 autocast over fp32 weights (Appendix A). The epoch was chosen on the train slice. The three main configurations used seeds 0–9; a seed fixes the head initialisation and data order, so seeds are paired across configurations.

What improved the model. Two changes improved single models (Table 2, Figure 1). Training for 16 epochs instead of 8 raised S2 by +0.047 ($t = 3.4$, 9/10 seeds up) and S1 by +0.026 ($t = 2.7$, 8/10). Adding the full question at 16 epochs gave a further +0.022 S2 ($t = 2.0$, 7/10) and +0.012 S1 ($t = 1.3$, 6/10). Together the two changes add +0.069 S2 ($t = 3.6$) and +0.038 S1 ($t = 2.8$), each on 9 of 10 seeds. Most of the gain comes from

Configuration	Single model		System	
	S2	S1	S2	S1
Baseline, 8 ep.	0.315 ± 0.040	0.576 ± 0.030	0.428	0.601
+ 16 epochs	0.362 ± 0.026	0.601 ± 0.016	0.400	0.620
+ full question	0.384 ± 0.030	0.614 ± 0.027	0.405	0.648
All of train [†]	0.416 ± 0.020	0.614 ± 0.016	0.489	–

Table 2: DeBERTa-v3-large on dev. Single model: mean $\pm$ sd over seeds 0–9, argmax, no dev tuning. System: the 10-seed mean probability with logit adjustment, τ fitted by 5-fold nested CV. Each of rows 2–3 adds one change to the row above; row 3 is the final system. [†]Full question, 16 epochs, all of train, last epoch; seeds 0–4 only, so its system is a 5-seed number and is not comparable with the others.

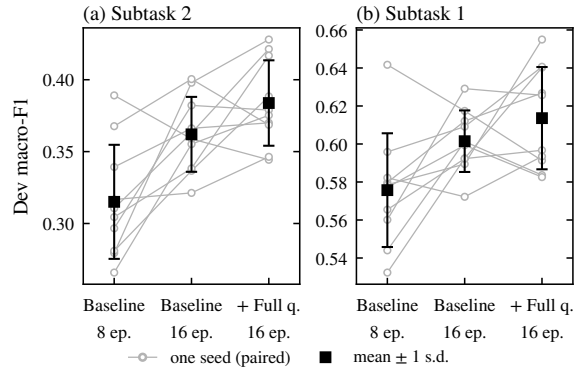


Figure 1: Single DeBERTa-v3-large models on dev for the three configurations: (a) S2 multi-reference macro-F1, (b) S1 macro-F1. Open circles joined by thin lines are individual seeds, paired across configurations; filled squares are the mean over the 10 seeds, with error bars of ± 1 standard deviation.

longer training, which also lifted the classes the baseline handled worst: General rose from 0.119 to 0.295 per model (seeds 0–4).

Undertraining at first looked like a negative result. Our first test added the full question together with Balanced Softmax and focal loss (Ren et al., 2020; Lin et al., 2017) at 8 epochs, lost to the baseline, and was blamed on the loss. A one-change control with plain cross-entropy at 8 epochs scored lower still: 0.282 ± 0.043 against 0.337 ± 0.044 on seeds 0–4. Its final training loss (1.32–1.50, against 0.57–1.09) showed that it had not finished fitting; its last epoch sat where the baseline had been at epoch 3. Its seeds were less confident (mean top probability 0.445 against 0.594) and leaned on the class prior. After logit adjustment, single models matched the baseline (0.353 against 0.357). At 16 epochs the same input gave the best single model. The deficit came from the training budget, not from the in-

put or the loss, so every run now reports its final training loss and selected epoch next to its scores.

From model to system. A system averages the ten seeds’ probabilities and applies logit adjustment (Eq. 2). On S2, better single models did not give a better system. Ensembling helped the final model more (0.384 to 0.409) than the baseline (0.315 to 0.319). Logit adjustment then lifted the baseline ensemble by +0.109 (to 0.428) and the final ensemble not at all (0.409 to 0.405; $\tau = 0.4$ when fitted on all of dev). The final system scores 0.405 S2 and 0.648 S1, against 0.428 and 0.601 for the baseline system. The S2 difference is -0.022 , with a 95% bootstrap interval of $[-0.101, +0.061]$ over dev items (seeds fixed), so the two systems cannot be told apart on S2; on S1 the final system is ahead. Our reading (inferred, not tested directly) is that longer training and logit adjustment correct the same fault, the pull of a prior-leaning model towards frequent classes. Either one is enough, and together they add nothing. The final system was chosen by a rule written before the replication runs.

Five-seed systems are unreliable on 308 items. The baseline system scored 0.438 on seeds 0–4 and 0.356 on seeds 5–9; the final system scored 0.478 and 0.434. This spread between two draws of the same system is larger than most effects we wanted to measure. All system claims therefore use ten seeds, and the conclusions rest on per-seed paired comparisons.

One later variant helped. Training on all of train for 16 epochs and keeping the last epoch is supported by the fact that, on the ten full-question seeds, the last epoch already scored at least as well as the selected one (S2 $+0.011 \pm 0.020$, 7/10). Paired with the full-question model on seeds 0–4, it gained +0.029 S2 per model (4/5 seeds up) and lost 0.017 S1 (1/5); its 5-seed system scored 0.489. Given the spread above, this is a 5-seed result and not a ranking; a 10-seed run is proposed.

What did not help. Seven further ideas, each tested against a named control, left the model or the system where it was (details in Appendix A, Table 6). A second encoder that re-ranked the baseline’s top-3 labels from their definitions scored 0.354, against 0.365 for the baseline ensemble it re-ranked. Learning what each label means from 3,448 examples is harder than classifying them. Balanced Softmax with focal loss over-corrected towards rare classes (Explicit F1 fell from 0.679 to 0.319) and used up the gain that logit adjustment

would have given. Three hierarchy designs did not beat the flat model: post-hoc routing through the clarity level, a factorised output head, and a Non-Reply gate with branch specialists. The flat softmax already scores all nine labels jointly. Boundary experts for the three most-confused pairs changed S2 by +0.002, because they seldom overturn a decision the flat model makes from the same input. A uniform model soup (Wortsman et al., 2022) of the ten models scored 0.267. Our seeds differ in the random head initialisation and the data order, so their weights do not average. Nine per-class decision weights overfit 308 items, and a temperature with nine biases, fitted on one half of dev and scored on the other, swung between 0.29 and 0.46. An ensemble that mixed both inputs scored 0.403, against 0.405. In short, adding encoders or rules trained on the same 3,448 examples did not move the system. Section 5 tests the next candidate, the backbone itself.

5 Is the Gap Knowledge? An 8B LLM Classifier

The encoder track (§4) and the error analysis (§6) show that the model, not label noise, is what fails: many errors fall on items all three annotators agreed on, and a held-out annotator scores far above the model. Two explanations remain. The encoder may lack *knowledge*, the world and language knowledge a 0.4B-parameter encoder has not acquired, or it may lack *label meaning*, what each evasion type means, which a classifier must infer from about 3,400 examples. A larger pretrained classifier tests the first explanation with nothing else changed. Before any run we registered three predictions in the experiment log: the gain passes the screening bar (+0.015 on S2 with at least 2 of 3 seeds up); if knowledge is the gap, the single-model S2 mean lies in 0.43–0.50 and S1 rises by +0.02 to +0.05; and the gain falls mostly on the commitment boundary (Explicit $\leftrightarrow$ Implicit, General $\leftrightarrow$ Implicit/Explicit) and more on agreed than on contested items.

One change: the backbone. We took the best encoder configuration (sub-question, full question and answer, 16 epochs) and replaced DeBERTa-v3-large (He et al., 2023) with Qwen3-8B-Base (Qwen Team, 2025), adapted with LoRA (Hu et al., 2022) of rank 16 ($\alpha = 32$, dropout 0.05) on all linear projections of the attention and feed-forward blocks. A new 9-way linear head, trained in full, reads the

hidden state of the last token. The input text and budgets are those of the encoder: “Sub-question: ... Full question: ...” capped at 256 tokens, then the answer, right-truncated to fit 1,024 tokens in total. One fixed closing question (“How does the answer respond to the sub-question?”) is appended, so the token the head reads sits at the same position in every item, after the whole question and answer. Sequences are right-padded, and each run first checks that an item’s logits are the same alone and inside a padded batch, aborting otherwise; this guards against a head that reads a padding token. Training ran in bf16 with AdamW at a learning rate of 10^{-4} , a cosine schedule with 5% warmup, 3 epochs and an effective batch of 16 (micro-batch $4 \times$ accumulation 4). The learning rate and epoch count change because the backbone does. Everything else is identical: the same 3,103 training rows, the same 345-row train slice (asserted identical in code), plain cross-entropy, and the epoch with the best slice macro-F1; dev is never used for a choice. Seeds 0–9 are paired with the DeBERTa runs of the same seeds. Appendix B lists all hyperparameters and the compute.

Results at ten paired seeds. The 3-seed screen passed the bar (+0.081 on S2, 3 of 3 seeds up), so the run was extended to ten seeds; its mean of 0.462 ± 0.080 is superseded by the ten-seed numbers in Table 3. Per model, Qwen improves S2 by +0.092 ($t = 7.12$) and S1 by +0.095 ($t = 8.48$) on 10 of 10 seeds, more than the whole encoder track gained from its baseline (0.315) to its best model (0.384). As a system it reaches 0.543 / 0.746 against 0.405 / 0.648 (S2 difference +0.136, 95% bootstrap interval [+0.054, +0.224]), and the ranking holds on the 2-annotator reference sets (0.524 against 0.382). Logit adjustment (Menon et al., 2021) matters far more for Qwen (+0.062 per model, against +0.005 for DeBERTa); §6 traces this to a skewed class prior.

All registered size predictions held: S2 lies in the predicted 0.43–0.50 (the S1 gain exceeded its range), the ten-seed mean stayed within 0.42–0.48, Qwen beat DeBERTa on 10 of 10 seeds (at least 7 predicted), and the system gain exceeded +0.03. The prediction about *where* the gain falls was refuted (§6). Among published dev results (Figure 2), the single-pass system lies above TeleAI’s fine-tuned Qwen2.5-7B (0.495; Shi et al., 2026) and ChulaNLP’s encoder+LLM pipeline (0.52; Peerachaidecho and Rutherford, 2026) and below

TeleAI’s multi-call pipeline (0.617) and the human reference (0.684); these come from other protocols, so the comparison locates our system rather than ranking it.

Two single-change follow-ups. In the 3-epoch runs the slice F1 rose at every epoch and the last epoch was always selected. Under a cosine schedule this proves nothing, because the schedule itself favours the last epoch, where the learning rate has decayed; only a longer schedule tests undertraining. With 12 epochs (seeds 0–2, schedule stretched accordingly) the slice selected epochs 9, 11 and 8, and the slice F1 rose by +0.084 on 3 of 3 seeds, so the rule registered in advance adopted the longer schedule. Training loss fell to about 0.02 without the slice F1 declining. Dev S2 was 0.543 ± 0.015 (S1 0.742 ± 0.008), reported only; a ten-seed 12-epoch system is proposed, not yet run. Training on all 3,448 rows and keeping the last of 3 epochs gave $+0.017 \pm 0.045$ per model (6 of 10 seeds up) and a system of 0.575, but the system difference of +0.028 has a 95% interval of $[-0.026, +0.092]$ and S1 fell from 0.746 to 0.711. As with the encoder (§4), the 3-seed screen (+0.039) overstated a gain that ten seeds cannot distinguish from zero.

6 Analysis

This section asks where the two classifiers fail and what the 8B model changes. The error analysis uses the final DeBERTa system’s 10-seed ensemble with argmax and no dev tuning (§4); the comparison uses the paired 10-seed runs of §5. All figures are on dev.

The model, not label noise, fails. Of the ensemble’s 308 predictions, 137 (44.5%) fell outside the item’s reference set, and 53 of these were on items where all three annotators agreed. Annotator disagreement in this task is real and informative (Uma et al., 2021), but it does not account for the gap. Scored as the test set is scored, each annotator against the other two reaches 0.643–0.766 on S2 (mean 0.684); the same ensemble, on the same three two-annotator reference sets, reaches 0.381, 0.380 and 0.381.

Commitment, not only topic. By TeleAI’s definition (an error that involves Dodging, General or Deflection in the prediction or the reference set), 86.9% of the ensemble’s errors are “triangle” errors, against 77.7% for their system (Shi et al., 2026). Only 29.2% of the errors, however, have

dev	Treatment	Control	Δ	Seeds up	95% CI
<i>Qwen3-8B-Base + LoRA (treatment) vs. DeBERTa-v3-large, full question, 16 epochs (control); 10 paired seeds</i>					
Single model S2	0.476 $\pm$ 0.043	0.384 $\pm$ 0.030	+0.092	10/10	
Single model S1	0.709 $\pm$ 0.031	0.614 $\pm$ 0.027	+0.095	10/10	
Single model + LA, S2	0.538 $\pm$ 0.043	0.389 $\pm$ 0.019	+0.149	10/10	
Ensemble (argmax), S2	0.495	0.409			
System S2	0.543 (0.549)	0.405 (0.412)	+0.136		[+0.054, +0.224]
System S1	0.746	0.648	+0.098		
System S2, 2-annotator sets	0.524	0.382	+0.142		
<i>Follow-ups: one change each (treatment) vs. Qwen, 3 epochs (control), same seeds</i>					
12 epochs, single model S2 (seeds 0–2)	0.543 $\pm$ 0.015	0.462 $\pm$ 0.080	+0.082 [†]		
All of train, single model S2	0.492 $\pm$ 0.045	0.476 $\pm$ 0.043	+0.017	6/10	
All of train, system S2	0.575 (0.562)	0.543 (0.549)	+0.028		[−0.026, +0.092]

Table 3: Qwen3-8B-Base with LoRA against the best DeBERTa-v3-large configuration on dev (multi-reference macro-F1). Single model: one seed, argmax, no dev tuning, mean $\pm$ s.d. over seeds; Δ is the paired mean difference. LA: logit adjustment. Ensemble: mean of the 10 seeds’ probabilities. System: ensemble + LA with τ by 5-fold nested CV, in parentheses the mean over 10 CV splits; where a CI is given, Δ and the CI come from a bootstrap over dev items, and the other system Δ s are plain differences. 2-annotator sets: mean over the three 2-annotator reference sets (the test regime). [†]Reported only; the 12-epoch schedule was adopted on the train slice (+0.084, 3/3 seeds).

both the prediction and a reference label inside the triangle. The largest error pairs (reference label $\rightarrow$ prediction) are Implicit $\rightarrow$ Explicit (25), General $\rightarrow$ Implicit (20) and General $\rightarrow$ Explicit (20): they concern how committed an answer is rather than whether it stays on topic. Length also matters. The 20 inputs still cut at 1,024 tokens are in-set only 20% of the time (0.580 for the other 288), and the shorter half of dev scores 0.400 on S2 against 0.283 for the longer half. Test inputs (sub-question, question and answer together) are much shorter (median 137 tokens against 474 on dev), so we infer, without being able to measure it, that dev understates test performance on this axis; the two-annotator test references cut the other way.

A typical error shows the knowledge an answer can depend on. Asked “*Would you campaign against Senator Joe Lieberman on Iraq?*”, the speaker replied “*I’m going to stay out of Connecticut.*” The annotators gave Dodging, Dodging and Implicit; the ensemble predicted General ($p = 0.37$). Our reading, which is an inference rather than a measurement, is that recognising a near-answer here requires knowing that Lieberman is a senator from Connecticut. Errors of this kind motivated the knowledge hypothesis tested with the 8B classifier.

Where the 8B classifier gains. Before the runs, we predicted that Qwen’s gain would fall mostly on the commitment boundary (Explicit $\leftrightarrow$ Implicit, General $\leftrightarrow$ Implicit/Explicit) and more on agreed items than on contested ones. Table 4 refutes both

parts. Per-class F1 (single models, mean over the 10 paired seeds) rises in 7 of 9 classes, most in Claims ignorance (+0.403), Declining (+0.220), Dodging (+0.103) and Deflection (+0.089). The commitment classes move least: General +0.059, Explicit +0.057, Implicit +0.040. Partial/half-answer is never predicted by either model, and Clarification falls by 0.144 on only 4 dev items. The in-set rate rises by +0.076 on unanimous items, +0.105 on items with two distinct labels and +0.048 on items with three, so the gain is not larger on agreed items. The largest gains are on the Non-Reply classes, which turn on recognising what kind of statement an answer is (a claim of ignorance, a refusal); General remains the weakest frequent class (F1 0.322).

Ranking, calibration and the decision rule.

Both models rank better than they decide. For Qwen (seeds 0–2), an acceptable label is the top prediction for 60.4% of items, in the top two for 83.1%, in the top three for 93.1% and in the top five for 99.1%. Confidence tracks correctness: across fifths of dev from least to most confident, the in-set rate rises 0.40, 0.48, 0.56, 0.70, 0.87. The DeBERTa ensemble shows the same pattern at a lower level, with an acceptable label in its top three for 93.8% of items and an in-set rate from 0.435 in the least confident fifth to 0.823 in the most confident. Most remaining errors are therefore a choice among a few candidates the model already ranks highly.

Qwen’s raw predictions are also skewed towards

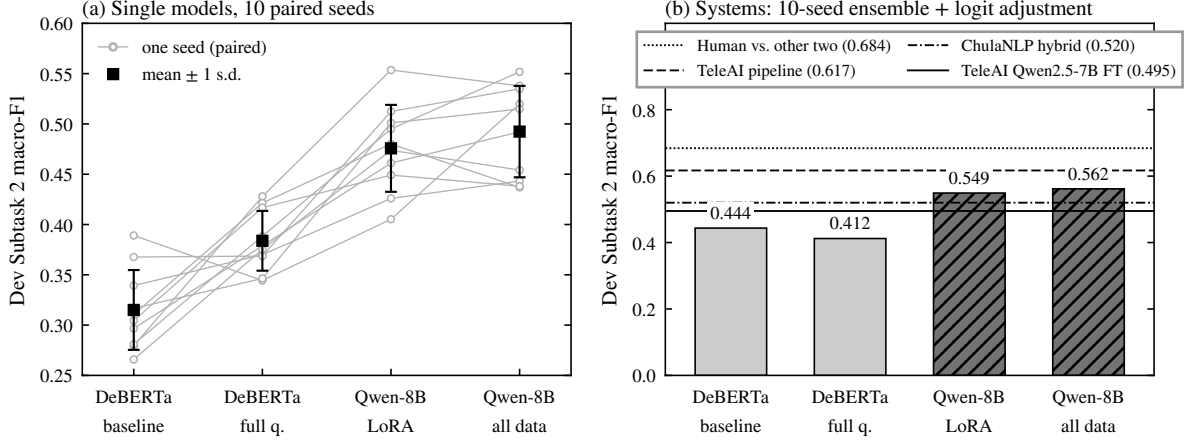


Figure 2: Dev S2 multi-reference macro-F1 of the DeBERTa-v3-large baseline, DeBERTa with the full question and 16 epochs, Qwen3-8B-Base with LoRA on the same input (3 epochs), and Qwen trained on all of train. (a) Single models on 10 paired seeds: open circles joined by thin lines are individual seeds; filled squares are means with ± 1 standard deviation. (b) The corresponding 10-seed systems (ensemble + logit adjustment, τ by nested CV; bar height is the mean over 10 CV splits), with published dev results as horizontal lines: TeleAI’s multi-call pipeline (0.617) and fine-tuned Qwen2.5-7B (0.495) (Shi et al., 2026), ChulaNLP’s DeBERTa top-5 + Kimi-K2 pipeline (0.52) (Peerachaidecho and Rutherford, 2026), and a human annotator scored against the other two (0.684).

	DeBERTa	Qwen	Δ
<i>per-class F1</i>			
Explicit	0.672	0.729	+0.057
Implicit	0.403	0.443	+0.040
Dodging	0.484	0.587	+0.103
General	0.263	0.322	+0.059
Deflection	0.287	0.377	+0.089
Partial/half-answer	0.000	0.000	+0.000
Declining to answer	0.381	0.601	+0.220
Claims ignorance	0.296	0.700	+0.403
Clarification	0.667	0.524	-0.144
<i>in-set rate by distinct annotator labels</i>			
one ($n = 125$)	0.529	0.605	+0.076
two ($n = 150$)	0.493	0.598	+0.105
three ($n = 33$)	0.688	0.736	+0.048

Table 4: Where the backbone gain falls: DeBERTa-v3-large (full question, 16 epochs) and Qwen3-8B-Base with LoRA, single models, mean over the same 10 seeds, dev S2 (multi-reference). The pre-registered prediction of a gain concentrated on the commitment boundary and on agreed items is refuted.

frequent classes. On seeds 0–2 it predicted General for only 24 items, although General is in 113 reference sets, while it predicted Explicit 111 times against 115. This is consistent with the effect of logit adjustment (Menon et al., 2021), which corrects the class prior with a single fitted parameter: it adds +0.062 per Qwen model (0.476 to 0.538) but only +0.005 per DeBERTa model (0.384 to 0.389). Finally, the two models do not complement each other. A probability mix, with the weight and τ fitted inside the nested cross-validation, scores 0.543

against 0.549 for Qwen alone (both means over 10 CV splits), and the folds place a mean weight of 0.79 on Qwen. The system-level comparison is in Table 3.

7 Conclusion and Next Steps

A carefully controlled encoder for CLARITY stops at 0.384 Subtask 2 macro-F1 per model (0.405 as a system), and the evidence places the limit in the model rather than in the labels or the decision rule (§4, §6). Changing only the backbone to a LoRA-tuned 8B decoder improves both subtasks on all ten paired seeds and lifts the system to 0.543 / 0.746 (§5), still below the best multi-call pipeline (0.617) and the human reference (0.684). The gain is broad rather than concentrated where we predicted, which suggests that a larger pretrained model reads better what kind of response an answer is, rather than resolving one difficult boundary.

Proposed next steps (not yet run): the twelve-epoch schedule on ten seeds; a cascade that tests the remaining explanation, label meaning, by sending only uncertain items with the classifier’s top candidates to a larger LLM given label definitions and boundary examples (Shi et al., 2026), traced as accuracy against LLM calls per item; and distillation of that cascade back into the single-pass model (Hinton et al., 2015), with every point measured in cost per item.

Limitations

Development-set evaluation. All results are on the 308-item development set, because the test labels were never released and the evaluation server has closed. The development set has three annotators per item and the test set two; we re-score with every two-annotator subset to approximate the test regime, but cannot measure the shift in answer length between the splits. With 308 items, system-level differences carry wide intervals: the gain of the 8B system has a 95% bootstrap interval of [0.054, 0.224], and the all-data variant could not be separated from its control.

Seeds and selection. Seed variance is large (the standard deviation of single-model Subtask 2 scores is 0.043 for the 8B classifier), so claims rest on ten paired seeds; the twelve-epoch schedule was screened on three seeds only and is adopted, not established. Epochs are chosen on a 345-row slice of the training data, which is itself noisy. The one decision-rule parameter is fitted by nested cross-validation on the development set, which keeps the score held out but means the system uses development labels to fit one scalar.

Scope of the comparison. The backbone change also changes the training recipe that a backbone requires (learning rate, epochs, parameter-efficient fine-tuning), so “knowledge” is shorthand for everything that comes with a larger pretrained model. We cannot rule out that the pretraining data of the 8B model contains the public interview transcripts the dataset is drawn from. Published comparison points come from other teams’ papers and protocols and are not directly controlled.

Coverage. The rarest class (*Partial/half-answer*, 14 development reference sets) is never predicted by any of our models, and the study covers one dataset in one language (English, U.S. presidential interviews).

Ethical Considerations

The models classify the *form* of an answer relative to a question; they do not judge honesty, intent or the truth of what is said, and a prediction such as *Dodging* should not be presented as a verdict on a speaker. Annotators themselves disagree on many items (§6), and automated labels of political speech can be misused to support one-sided narratives; any deployment should report uncertainty

and keep humans in the loop. The data are public interview transcripts released for the shared task. GPU experiments for the 8B classifier ran on rented single-GPU machines (up to four at once, for about two hours of wall-clock time); Appendix B reports their cost.

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References

- Peter Bull. 1994. On identifying questions, replies, and non-replies in political interviews. *Journal of Language and Social Psychology*, 13(2):115–131.
- Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman & Hall, New York.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *The Tenth International Conference on Learning Representations (ICLR)*.
- Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988.
- Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *The Seventh International Conference on Learning Representations (ICLR)*.
- Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. 2021. Long-tail learning via logit adjustment. In *The Ninth International Conference on Learning Representations (ICLR)*.

Wisarat Peerachaidecho and Attapol T. Rutherford. 2026. ChulaNLP at SemEval-2026 Task 6: A hybrid BERT–LLM framework for political response clarity and evasion detection. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, pages 1874–1881. Association for Computational Linguistics.

Barbara Plank, Dirk Hovy, and Anders Søgaard. 2014. Linguistically debatable or just plain wrong? In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 507–511, Baltimore, Maryland. Association for Computational Linguistics.

Qwen Team. 2025. [Qwen3 technical report](#). Preprint, arXiv:2505.09388.

Jiawei Ren, Cunjun Yu, Shunan Sheng, Xiao Ma, Haiyu Zhao, Shuai Yi, and Hongsheng Li. 2020. Balanced meta-softmax for long-tailed visual recognition. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*.

Lingling Shi, Haoyu Jin, Shiquan Wang, Ruiyu Fang, Shuangyong Song, and Xuelong Li. 2026. TeleAI at SemEval-2026 Task 6: A confidence-aware multi-stage reasoning framework with chain-of-thought. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, pages 1591–1609. Association for Computational Linguistics.

Konstantinos Thomas, Giorgos Filandrianos, Maria Lymperaïou, Chrysoula Zerva, and Giorgos Stamou. 2024. “I never said that”: A dataset, taxonomy and baselines on response clarity classification. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 5204–5233, Miami, Florida, USA. Association for Computational Linguistics.

Konstantinos Thomas, Giorgos Filandrianos, Maria Lymperaïou, Chrysoula Zerva, and Giorgos Stamou. 2026. [SemEval-2026 Task 6: CLARITY – unmasking political question evasions](#). Preprint, arXiv:2603.14027.

Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35 (NeurIPS)*.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing.

In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. 2022. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*.

A Encoder-track details

Hyperparameters. Table 5 lists the settings, taken from `clarity/encoder.py`, `clarity/decide.py` and the experiment argument files. Baseline and final models differ only in the input and the number of epochs. One implementation detail mattered: transformers 5.x loads a checkpoint in its stored dtype, and DeBERTa-v3’s published weights are fp16, in which its attention overflows. With gradient clipping masking the overflow, the model trained smoothly to the label prior (loss 1.887, the label entropy) and predicted Explicit for every item. We therefore load the weights in fp32 explicitly.

Negative results. Table 6 lists each idea that did not help, with its control and the reason the experiment log gives. Rows marked “5 seeds” compare against the baseline’s or the full-question model’s seeds 0–4. The baseline’s 5-seed system (0.438) is the favourable draw discussed in §4.

B LLM-classifier details

Table 7 lists the configuration of the Qwen3-8B-Base classifier of §5; everything not listed matches the best encoder configuration. The training script is `clarity/llm_classifier.py`.

Compute. All LLM runs used JarvisLabs virtual machines with one NVIDIA RTX PRO 6000 (96 GB) each; the ten-seed round (seeds 3–9, the 12-epoch runs and the all-of-train runs) used four machines in parallel. Gradient checkpointing was on for the 3-epoch screen (seeds 0–2) and off for every later run. It changes speed and memory, not the computation: the timing pilot measured 2.11 s per optimizer step with checkpointing and 1.43 s without, and training peaked at 19.3 and 53.4 GiB of GPU memory respectively. A 3-epoch seed took 21 min with checkpointing and 15–16 min without; a 12-epoch seed took about 59 min. The ten-seed

Setting	Value
Backbone	microsoft/deberta-v3-large, all weights trained
Head	[CLS] $\rightarrow$ dense + tanh $\rightarrow$ dropout 0.1 $\rightarrow$ linear (9)
Input, baseline	sub-question (≤ 96 tokens) [SEP] answer; 512 tokens
Input, final	“Sub-question: ... Full question: ...” (≤ 256 tokens) [SEP] answer; 1,024 tokens
Truncation	answer cut from the right
Loss	cross-entropy, no smoothing, no class weights
Optimiser	AdamW, $\beta = (0.9, 0.98)$, $\epsilon = 10^{-6}$, weight decay 0.01 (none on biases and LayerNorm)
Learning rate	$10^{-5} \times 0.95^k$ for a layer k levels below the head; head and pooler 10^{-4}
Schedule	cosine, warm-up 10% of steps
Batch	16, no accumulation; gradient-norm clipping 1.0
Epochs	8 (baseline) or 16
Precision	fp32 weights, bf16 autocast; gradient checkpointing
Epoch selection	best macro-F1 on the 345-row train slice; last epoch for all-of-train runs 0–9 (all of train: 0–4)
Seeds	0–9 (all of train: 0–4)
Logit adjustment	$\tau \in [-1, 2]$, 61-point grid; prior = train label frequencies; 5-fold nested CV on dev
Hardware	one shared RTX PRO 6000 (96 GB); peak memory about 9 GB at 512 tokens, 15–17 GB at 1,024

Table 5: DeBERTa-v3-large training and decision settings.

round cost INR 1,518.79 of GPU time, read from the provider’s account balance before and after.

Pipeline and logging. Each machine ran the same scripted pipeline (`jarvis_setup.sh`, driven over SSH by `jarvis_drive.sh`): environment setup and model download; a smoke test with Qwen3-0.6B-Base that runs the padding check and interrupts and resumes an epoch; a 20-step timing pilot that measures peak memory on the longest micro-batch and aborts if the run is projected over a time budget; training with a resume checkpoint after every epoch; and a pause of the machine when done or on failure. From the ten-seed round on, training metrics were logged to Weights & Biases every 20 optimizer steps (loss, learning rate, gradient norm, throughput) and after every epoch (slice F1, and dev scores that were logged but never used for a choice), the LoRA adapter of every epoch was kept, and each finished seed was uploaded to a Hugging Face repository as soon as it ended. Every seed’s probabilities, per-epoch history and selected adapter are in that repository.

C Reproducibility

Code and records. All code, experiment definitions and the experiment log live in the project repository. Every number in this paper traces to a file: `midieval/RESULTS.md` lists each number with its source, the chronological experiment log (`clarity/reports/02_experiment_log.md`) holds the pre-registered hypotheses and predictions written before each run, and the raw analysis outputs are in `clarity/reports/raw/`. The encoder analyses are regenerated by `clarity/mideval_analysis.py` and `clarity/e11_replication.py`, and the 8B-classifier analyses and Figure 2 by `clarity/e13_analysis.py` and `clarity/e13_figures.py`; all run on CPU from the saved per-seed probability matrices.

Trained runs. For every seed of every configuration, the development, train-slice and test probability matrices, the per-epoch history and the selected weights (LoRA adapters for the 8B classifier) are stored in a Hugging Face model repository under `<configuration>/seed<k>/`; per-step and per-epoch training curves are logged to Weights & Biases.

Software. Python 3.12, PyTorch 2.11 (CUDA 12.8), Transformers 5.17 (Wolf et al., 2020), PEFT 0.20 and Accelerate 1.15; model weights are loaded with an explicit dtype (Appendix A).

Hardware. Encoder runs used a shared NVIDIA RTX PRO 6000 (96 GB); 8B-classifier runs used rented virtual machines with one RTX PRO 6000 each, driven by two shell scripts that build the environment, run a smoke test and a timing pilot that aborts runs projected over a time budget, train, upload results and pause the machine (Appendix B).

Idea	Result (dev S2)	Control	Why it did not help
Second DeBERTa re-ranks the baseline’s top-3 labels from their definitions (5 seeds)	0.354 ± 0.003 with the baseline prior; 0.329 alone	0.365 (baseline ensemble, argmax)	Training loss moved only $1.06 \rightarrow 0.88$ in 8 epochs ($1.10 = \text{chance among three}$); learning what a label means from its definition is harder than classifying. It fixed 25 baseline errors and broke 37 correct answers.
Full question + Balanced Softmax + focal loss, 8 epochs (5 seeds)	system 0.379; single 0.322 ± 0.021	0.438; 0.337 ± 0.044	Over-corrected towards rare classes (Explicit $0.679 \rightarrow 0.319$, Dodging $0.475 \rightarrow 0.081$) and used up the gain of logit adjustment; most of the deficit was the undertrained input.
Post-hoc routing: clarity level first, then leaf	0.369	0.365 (ensemble, argmax)	The model’s own top 3 contains an acceptable label for 89.6% of items, against 81.2% for the taxonomy branch (3.60 labels).
Factorised head $p(\text{level}) p(\text{leaf} \mid \text{level})$ (5 seeds)	system 0.370; single 0.320 ± 0.038	0.438; 0.337 ± 0.044	S1 steadier across seeds (sd 0.010 against 0.025 for the level read-out) but not better; S2 lower.
Non-Reply gate + branch specialist encoders	3-seed system 0.390; flat model’s gate + specialists -0.009 per model with the rule (5 seeds, 2/5)	0.418 (flat, 3 seeds)	The gate memorises the binary split (training loss ≈ 0.03), so soft routing becomes hard. The specialists passed the 3-seed screen and were level on 5 seeds.
Boundary experts for Implicit/Dodging, Explicit/Implicit and General/Deflection (3 seeds)	$+0.002$ per model (1/3); S1 -0.004	flat model, same seeds	They seldom overturn a decision the flat model makes between the same two labels from the same input.
Uniform model soup of the 10 full-question models	0.267 (greedy soup kept 1 model: 0.381)	0.384 (single model)	Seeds differ in the random head initialisation and the data order, so their weights do not lie in one basin; averaging any two already lowered the slice score.
Nine per-class multipliers, nested CV (5-seed baseline ensemble)	0.394 (in-fold minus held-out gap $+0.098$)	0.438 (logit adjustment)	Nine parameters overfit 308 items; below argmax on two of the three 2-annotator reference sets.
Temperature + nine biases, fitted on one dev half and scored on the other	0.293–0.461 over 4 configurations $\times 2$ directions	logit adjustment	Erratic: the same rule swings by up to 0.148 between the two halves.
Mixed-input ensemble: full question and sub-question + answer, both 16 epochs, seeds 5–9	0.403	0.405 (10 full-question models)	The idea came from one set of seeds (0.498) and did not replicate on fresh seeds.

Table 6: Encoder-track ideas that did not help, all on dev S2. “System” is the seed ensemble with logit adjustment (τ by nested CV); “per model” is a mean of per-seed paired differences. Sources: clarity/reports/02_experiment_log.md (E4–E12) and mideval/RESULTS.md.

Setting	Value
Backbone	Qwen3-8B-Base, bf16
Adapter	LoRA, $r = 16$, $\alpha = 32$, dropout 0.05
LoRA targets	q, k, v, o, gate, up, down_proj
Head	new 9-way linear, last non-pad token, trained in full
Precision	fp32 trainable weights, bf16 autocast
Optimizer	AdamW, lr 10^{-4} , weight decay 0
Gradient clipping	norm 1.0
Schedule	cosine, linear warmup 5% of steps
Epochs	3 (12 in the follow-up)
Batch	4×4 accumulation = 16
Steps per epoch	194 (3,103 training rows)
Loss	cross-entropy, unweighted
Input	question part ≤ 256 tokens; total $\leq 1,024$ incl. closing question
Batching	length-sorted within chunks of 50 batches; order fixed by (seed, epoch)
Epoch selection	best macro-F1 on the 345-row slice (all-of-train runs: last epoch)
Padding check	abort if relative logit difference > 0.2
Seeds	0–9 (12 epochs: 0–2)

Table 7: Hyperparameters of the LLM classifier.